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Batch – S11
Roll no. – 07

Aim – Implementation of Hashing Function with different Collision Tech

Consider a hash table of size 10

Keys: 72, 24, 36, 24, 63, 81, 91, 100

Linear probing:

$$h(x,i) = (h(x) + i) \bmod 10$$

Insert 72 (Step 1).

$$\begin{aligned}(72,0) &= (72 + 0) \bmod 10 \\ &= 2\end{aligned}$$

Insert 27 (Step 2)

$$\begin{aligned}(27,0) &= (27 + 0) \bmod 10 \\ &= 7\end{aligned}$$

Insert 36 (Step 3)

$$\begin{aligned}(36,0) &= (36 + 0) \bmod 10 \\ &= 6\end{aligned}$$

Insert 24 (Step 4)

$$\begin{aligned}(24,0) &= (24 + 0) \bmod 10 \\ &= 4\end{aligned}$$

Insert 63 (Step 5)

$$(63,0) = (63 + 0) \bmod 10$$

0	100
1	81
2	72
3	63
4	24
5	91
6	36
7	27
8	

=3

Insert 81 (Step 6)

$$(81, 0) = (81 + 0) \bmod 10$$

=1

Insert 91 (Step 7)

$$(91, 0) = (91 + 0) \bmod 10$$

=1 (Collision)

Step 8;

$$(91, 2) = (91 + 1) \bmod 10$$

=27 (Collision)

Step 9:

$$(91, 2) = (91 + 2) \bmod 10$$

=3 (Collision)

Step 10:

$$(91, 3) = (91 + 3) \bmod 10$$

=4 (Collision)

Step 11:

$$(91, 4) = (91 + 4) \bmod 10$$

=5 (Collision)

Insert 100 (Step 12)

$$(100, 0) = (100 + 0) \bmod 10$$

=0

Quadratic Probing:

$$h(x, i) = (h(x) + i^2) \bmod 10$$

Insert 72 (Step 1)

$$(72, 0) = (72 + 0) \bmod 10$$

$$= 2$$

Insert 27

$$i=0, h(27,0) = 7$$

Insert 36

$$i=0, h(36,0) = 6$$

Insert 24

$$i=0, h(24,0) = 4$$

Insert 63

$$i=0, h(63,0) = 3$$

Insert 81

$$i=0, h(81,0) = 1$$

Insert 91

$$i=0, h(91,0) = 1 - \text{Collision}$$

$$i=0, h(91,1) = 2 - \text{Collision}$$

$$i=0, h(91,2) = 5$$

Insert 100

$$i=0, h(100,0) = 0$$

0	100
1	81
2	72
3	63
4	24
5	91
6	36
7	27
8	

Double Hashing:

$$h(x,p) = (h_1(x,i) + p h_2(x,i)) \bmod 10$$

$$h_2(x) = \text{prime} - (\text{key} \% \text{prime}) \ \& \ \text{prime} < \text{tables}$$

$$\text{hash 1 (72)} = 72 \% 10 = 2$$

$$\text{hash 1 (27)} = 27 \% 10 = 7$$

$$\text{hash 1 (36)} = 36 \% 10 = 6$$

$$\text{hash 1 (24)} = 24 \% 10 = 4$$

$$\text{hash 1 (63)} = 63 \% 10 = 3$$

$$\text{hash 1 (81)} = 81 \% 10 = 1$$

$$\text{hash 1 (91)} = 91 \% 10 = 1$$

$$\text{hash 2 (91)} = 7 - (91 \% 7)$$

$$= 7$$

$$[\text{hash}(91) + 1 * \text{hash}_2(91)] \% 10$$

$$= [1 + 7] \% 10$$

$$= 8$$

$$\text{Hash 1 (100)} = 100 \% 10 = 0$$

0	100
1	81
2	72
3	63
4	24
5	91
6	36
7	27
8	

Prime = 7

